

HealthParse: A Modular Pipeline for Privacy-Aware Extraction of Structured Clinical Information from Heterogeneous Healthcare Documents

PHASE I REPORT

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BONAFIDE CERTIFICATE

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“This project addresses the following Sustainable Development Goal(s): Goal 3: Good Health and Well-Being; Goal 9: Industry, Innovation and Infrastructure.”

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ABSTRACT

A hospital intake desk can receive a cleanly typeset laboratory PDF, a faxed discharge summary and a phone photograph of a handwritten prescription within the same hour, and all of them must become useful clinical data and not merely be stored as incomprehensible files. This project presents **HealthParse**, a modular document processing pipeline for such mixed document types. A document passes through four stages: ingestion and validation; format-aware text extraction, in which native-text PDFs are parsed directly while scanned or photographed documents are preprocessed and read with Tesseract OCR; privacy-aware de-identification of the recovered text using Microsoft Presidio with custom healthcare recognizers; and hybrid clinical information extraction that combines a zero-shot named-entity model (GLiNER) with deterministic rules for values such as dosages and laboratory results. The implementation retains the data structures that make the output auditable: per-word OCR confidence and bounding boxes, typed redaction placeholders, and character-level offsets linking every extracted entity to its source text. The pipeline was evaluated end to end on a six-document corpus of native-text PDFs and scanned images. All six documents were processed successfully, producing 164 structured clinical entities across eight categories and 484 identifier redactions across eleven identifier types, at a mean processing time of 5.83 seconds per document on a CPU-only machine. These are operational execution results and not an accuracy benchmark against labelled ground truth, and the report states this distinction, together with the limits of the system, explicitly.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CPU	Central Processing Unit
DPI	Dots Per Inch
GLiNER	Generalist Model for Named Entity Recognition Using Bidirectional Transformer
GPU	Graphics Processing Unit
JPEG	Joint Photographic Experts Group
JSON / JSONB	JavaScript Object Notation / binary JSON
MIME	Multipurpose Internet Mail Extensions
MRN	Medical Record Number
NER	Named Entity Recognition
OCR	Optical Character Recognition
PDF	Portable Document Format
PHI	Protected Health Information
PNG	Portable Network Graphics
RAM	Random Access Memory
REST	Representational State Transfer
UHID	Unique Health Identifier
UUID	Universally Unique Identifier

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Healthcare administrative offices rarely receive records in a single format. A laboratory returns a digitally generated PDF report with neatly embedded text. A referring clinic faxes a discharge summary that arrives as a low-quality, slightly warped scan. A patient photographs a handwritten prescription on a phone and uploads the picture through a portal. All three carry the same kind of information, yet only the first can be read by a machine directly. The other two need an intermediate step of optical character recognition (OCR) before any downstream system can use them, and OCR is far from a perfect process: its accuracy depends heavily on scan quality, lighting conditions, document skew and layout complexity.

Converting a document to text does not finish the job. The recovered text typically still contains identifying patient information, which must be handled appropriately before the content is used for its intended purpose. Doing all of this reliably, for documents in formats the system was not specifically trained on, is a significant engineering task in its own right, distinct from, and arguably harder than, training an extraction model.

This report describes **HealthParse**, a modular document processing pipeline built to handle exactly this task. Given a healthcare document in one of several common formats, the pipeline ingests and validates it, extracts any embedded machine-readable text with native PDF tools or performs OCR with Tesseract after an image-preprocessing sequence, de-identifies the extracted text, and finally extracts structured clinical information using a combination of zero-shot named-entity recognition and deterministic pattern extraction.

1.2 PROBLEM STATEMENT

A hospital intake desk may receive a cleanly typeset laboratory PDF, a faxed discharge summary and a phone photograph of a handwritten prescription within the same hour. All of them must become useful clinical data rather than merely be stored as incomprehensible files. The specific problems that arise are the following.

- **Format heterogeneity.** Native-text PDFs, scanned PDFs and camera images need different handling; running OCR on a document that already contains encoded text is wasteful, while skipping it on a scan yields nothing.
- **Variable text quality.** OCR output depends on scan quality, skew and noise, and its errors are inherited by every later stage.
- **Exposure of identifiable information.** Recovered text carries names, record numbers, dates and contact details

that must be reduced before clinical processing, while keeping the sentence structure that the extraction stage needs.

- **Structured, auditable extraction.** Free text and tabular values must be converted into labelled entities, and each entity should remain traceable to the place in the document it came from.
- **Integration.** Much of the literature studies one stage in isolation. Far less is said about how the stages fit into one running pipeline and which implementation choices that integration forces.

1.3 OBJECTIVES

- To design and implement a four-stage pipeline (ingestion, extraction, de-identification and clinical information extraction) that accepts PDF, JPEG and PNG healthcare documents.
- To handle native-text PDFs and scanned or photographed documents through two distinct extraction paths, and to preserve OCR layout information (per-word confidence and bounding boxes) instead of discarding it.
- To reduce exposure of identifiable information with a privacy layer built on Microsoft Presidio and custom healthcare recognizers, replacing identifiers with typed placeholders.

- To extract eight categories of clinical entity using a hybrid of a zero-shot NER model (GLiNER) and deterministic rules, retaining confidence scores and character-level provenance for every entity.
- To evaluate the implemented pipeline end to end on a mixed-format document set and to report, honestly and explicitly, what the evaluation does and does not establish.

1.4 SCOPE

This work is scoped deliberately to the chain from ingestion through clinical information extraction (modules M1 to M4). The following are within scope: document ingestion and validation; the dual native-PDF and OCR extraction paths with preprocessing; privacy-aware de-identification of extracted text; and hybrid clinical entity extraction with provenance tracking.

The following are explicitly outside the scope of this report: full table-structure recognition (dense laboratory grids are recovered as positioned text, not as tables); any guarantee that all identifying information has been removed; clinical decision support; and the platform's additional downstream capabilities, most notably document classification, which are neither evaluated nor claimed as contributions here.

1.5 CONTRIBUTIONS

- A description of the implemented pipeline at the level of concrete data structures and API responses, covering processing decisions that go beyond a conceptual architecture diagram.
- A detailed treatment of the dual processing path for native-text PDFs versus scanned or image documents, including what layout information is preserved during text extraction.
- A description of the privacy layer's capabilities and limitations, including an acknowledgement that automated de-identification can both over-redact and under-redact.
- A hybrid clinical extraction stage that combines zero-shot NER with deterministic rules across eight entity categories, with character-offset provenance linking each entity to a location in the source text.
- An end-to-end evaluation on a six-document test set mixing native PDFs and scanned images, reporting execution success rate, per-stage timing, entity counts by label and the privacy redactions applied.

1.6 ORGANIZATION OF THE REPORT

Chapter 2 reviews related work on OCR for healthcare documents, clinical named-entity recognition and privacy-preserving de-identification, and positions this work against it.

Chapter 3 presents the proposed system, its architecture and its requirements. Chapter 4 describes each module in detail. Chapter 5 covers the implementation and discusses the results of the end-to-end evaluation. Chapter 6 concludes the report and outlines future work. The references follow.

CHAPTER 2

RELATED WORKS

This chapter reviews existing work in the three areas on which the pipeline depends, namely OCR for healthcare documents, clinical named-entity recognition and de-identification of clinical text, and identifies the gap that this work addresses.

2.1 OCR FOR HEALTHCARE DOCUMENTS

OCR has been studied extensively for general digitization and for specific domains such as invoices and historical archives, but relatively little work has focused on healthcare-specific benchmarking. The literature agrees that OCR performance is not a static value. It varies with document quality, layout complexity and the degree of degradation in the scanned image, and errors introduced at this stage affect every subsequent text processing step [1]. Tesseract [1] remains a standard open-source baseline because of its maturity and low computational cost. In a benchmark on food-packaging labels, for example, Tesseract achieved the lowest character error rate among the engines compared on the ground-truth subset [2]; such rankings are known to shift with the document population, which is why this work does not attach accuracy figures from other studies to its own results.

Work on prescription information extraction has applied image processing steps such as resizing and adaptive

thresholding ahead of OCR to extract prescription information from document images [3]. Preprocessing is not uniformly beneficial, and quality-aware or adaptive enhancement stages have been proposed as an alternative to a fixed preprocessing sequence [4], which suggests that the preprocessing chain should be treated as a design decision to be revisited and not a fixed recipe.

2.2 CLINICAL NAMED ENTITY RECOGNITION

Extracting structured clinical entities from free text has traditionally relied on supervised models trained on annotated corpora from the clinical domain, such as Clinical BERT embeddings [5] and SciBERT [6]. More recent span-identification models such as GLiNER [7] allow the label set to be specified at inference time without retraining. This flexibility comes at the cost of some raw accuracy, but it lets a system adapt to new entity categories without fresh labelled data.

Deterministic, rule-based extraction remains common practice alongside learned models for structured value types such as laboratory measurements, dosages and dates. For such a narrow category a pattern-based approach can often give better results than a learned model and is easier to interpret and audit.

2.3 PRIVACY-PRESERVING DE-IDENTIFICATION

De-identification of clinical text is commonly approached as a mixture of rule-based recognizers for identifiers with a predictable form (phone numbers, dates, structured ID numbers) and statistical or learned recognizers for identifiers that appear in unstructured text, such as patient names [8]. Toolkits such as Microsoft Presidio [9] realize this mixture as a configurable pipeline of recognizers and anonymization operators, and are commonly extended with custom recognizers for domain-specific identifiers, for example medical record numbers in a healthcare setting.

It is well understood in this literature that no automated de-identification approach guarantees complete removal. False negatives (missed identifiers) and false positives (over-redaction of non-identifying text) are known failure modes and not edge cases. This report therefore describes its privacy layer as one that reduces exposure and makes no claim of guaranteed removal.

2.4 COMPARISON AND DISCUSSION

Table 2.1 summarises existing approaches and how this work addresses them.

Table 2.1: *Stage-wise summary of existing approaches and how this work addresses them*

Stage	Common approach	Typical limitation	Treatment in HealthParse
OCR	Single engine, fixed preprocessing, flat text output	Accuracy varies with scan quality; layout discarded	Two extraction paths; per-word confidence and bounding boxes retained
Clinical NER	Supervised models on annotated corpora	Needs labelled data for new categories	Zero-shot GLiNER plus deterministic rules
De-identification	Rule-based plus learned recognizers (e.g. Presidio)	No guarantee; false positives and negatives	Custom healthcare recognizers; typed placeholders; limits stated
Integration	Each stage studied in isolation	Little on how stages compose in practice	Single running pipeline with provenance end to end

The comparison shows that each individual stage is well studied. What is less documented is how the stages are chained: how OCR layout metadata is represented once extraction moves to a downstream stage, how de-identified text is passed forward while retaining document structure, and how an extracted entity keeps a traceable link back to its source.

2.5 RESEARCH GAP AND POSITIONING

Much of the existing literature works on one stage in isolation, such as an OCR engine comparison, and says far less about how the stages integrate into a single running pipeline. The contribution of this work lies at the level of integration. It is articulated concretely, with the data structures each stage produces, and demonstrated on a mixed-format document set and not argued in the abstract.

CHAPTER 3

PROPOSED SYSTEM

This chapter presents the overall design of the proposed **HealthParse** system, its architecture and data flow, the design principles that shaped it, and the software and hardware requirements of the implementation.

3.1 SYSTEM OVERVIEW

HealthParse is implemented as a layered application. A frontend client submits documents to a FastAPI backend, which orchestrates document ingestion, OCR and preprocessing, privacy de-identification and clinical information extraction. Results are persisted to a relational database together with the originally uploaded files and are exposed through versioned REST endpoints for uploading documents, retrieving text and retrieving entities.

From the perspective of a single document the system follows four stages: ingestion and registration (M1); document-type-specific text extraction (M2); privacy-aware de-identification (M3); and hybrid medical information extraction (M4). Each stage consumes the output of the previous one and adds structured information that later stages and the final user can inspect.

3.2 SYSTEM ARCHITECTURE

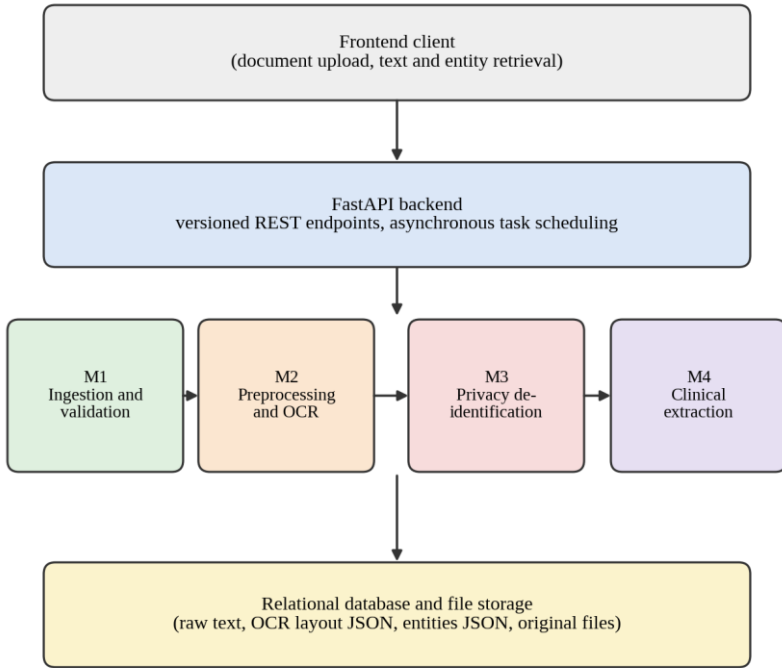


Figure 3.1: Layered architecture of HealthParse

As shown in Figure 3.1, the client layer is only responsible for submitting documents and retrieving results. The FastAPI backend accepts uploads, registers the document and schedules the remaining stages asynchronously, so the upload request returns without waiting for OCR and extraction. The four processing stages (M1, M2, M3 and M4) run behind the backend, and every artefact they produce (raw text, OCR layout, entities and the original file) is stored in the persistence layer.

3.3 DATA FLOW

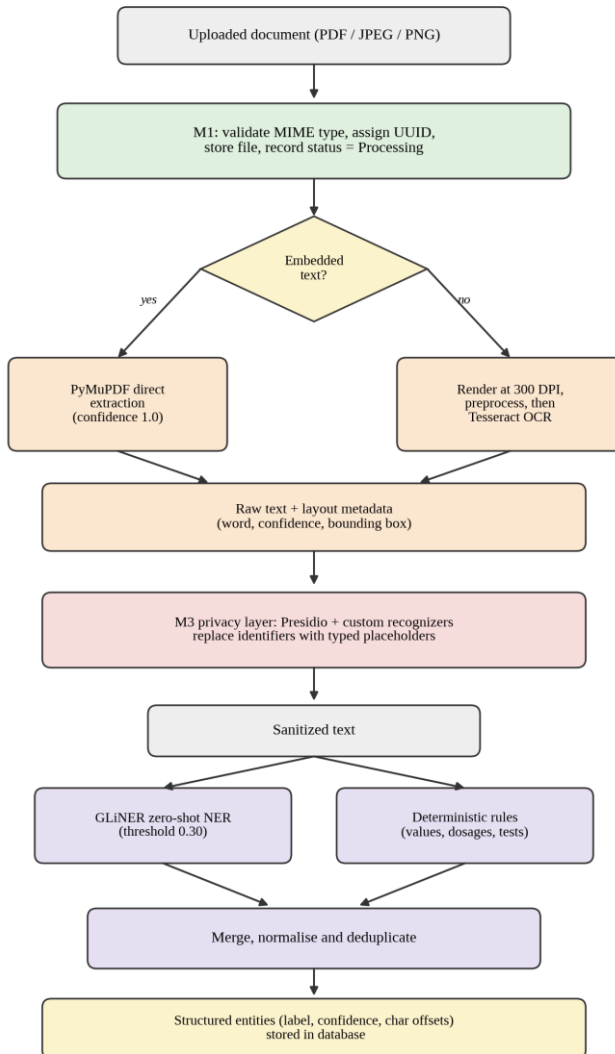


Figure 3.2: End-to-end data flow for a single document

Figure 3.2 follows a single document through the system. After validation, the pipeline decides between two extraction paths. A digitally generated PDF that already contains embedded text is parsed directly. A scanned PDF or an image is rendered, preprocessed and read by Tesseract. Both paths produce raw text with layout metadata, which is de-identified before any downstream processing touches it. The sanitized text is then processed in parallel by GLiNER and the deterministic rule engine, and the merged results are stored as structured entities.

3.4 DESIGN PRINCIPLES

- **Decoupled ingestion.** Ingestion latency is separated from the time needed for OCR and later stages by scheduling them asynchronously after the document record is created.
- **Format-aware extraction.** Native-text PDFs bypass OCR, because there is nothing to recognize visually when the text is already encoded in the file.
- **Nothing discarded early.** OCR output keeps per-word text, confidence and bounding boxes, so spatial provenance remains available after extraction completes.
- **Privacy before processing.** De-identification runs before any clinical processing, and identifiers are replaced with typed placeholders so that sentence structure is preserved.

- **Complementary extraction.** A learned model handles free-text concepts, and deterministic rules handle value types with consistent patterns, where matching is easier to audit.
- **Traceability.** Every stored entity keeps its label, a confidence score and character start and end offsets into the sanitized text.
- **Honest claims.** The system makes no claim of guaranteed removal of identifying information, and its evaluation is presented as operational execution evidence and not an accuracy benchmark.

3.5 SYSTEM REQUIREMENTS

3.5.1 SOFTWARE REQUIREMENTS

Table 3.1 lists the software components used.

Table 3.1: *Software requirements of the proposed system*

Component	Specification
Programming language	Python 3.10.0
Backend framework	FastAPI with versioned REST endpoints
Native PDF text extraction	PyMuPDF
Image preprocessing	OpenCV-based chain
OCR engine	Tesseract OCR 5.5.3 (Leptonica 1.87.0)
De-identification	Microsoft Presidio analyzer and anonymizer with custom healthcare recognizers
Clinical NER model	GLiNER, urchade/gliner_small-v2.1
Persistence	Relational database; layout and entities stored as JSON/JSONB

3.5.2 HARDWARE REQUIREMENTS

Table 3.2: *Hardware configuration used for the evaluation*

Component	Specification
Processor	12th Gen Intel Core i5-1235U
Memory	8 GB RAM
Graphics	Integrated Intel UHD Graphics
Accelerator	None; CPU-only configuration, no discrete GPU used

CHAPTER 4

MODULE DESCRIPTION

This chapter describes each module of the pipeline in detail, including the data structures it produces. Table 4.1 summarizes the modules before they are discussed one by one.

Table 4.1: *Summary of the pipeline modules*

Module	Input	Processing	Output
M1 Ingestion	PDF, JPEG or PNG upload	MIME validation, UUID assignment, storage, async scheduling	Document record with Processing status
M2 Extraction	Stored document	Native parsing or preprocessing plus Tesseract OCR	Raw text, layout JSON, confidence
M3 Privacy	Raw text	Presidio with custom recognizers, typed placeholders	Sanitized text
M4 Extraction	Sanitized text	GLiNER plus deterministic rules, merge and deduplication	Structured entities with offsets

4.1 DOCUMENT INGESTION (M1)

Documents are uploaded through a REST endpoint that accepts PDF, JPEG and PNG file types. Each uploaded file is validated against the MIME type declared for it. A UUID is then generated and associated with the uploading user and an optional patient identifier, and the file is written to storage. A document record is created with a Processing status, and the

remaining pipeline stages are scheduled asynchronously instead of being executed inline with the upload request.

This design decouples ingestion latency from the time required for OCR and the later stages. It is referred to as ingestion and registration, and not simply upload, because the behaviour affects how the rest of the pipeline is composed and reasoned about.

4.2 PREPROCESSING AND TEXT EXTRACTION (M2)

Once a document is queued, the system chooses between two extraction paths depending on document type, as shown in Figure 4.1.

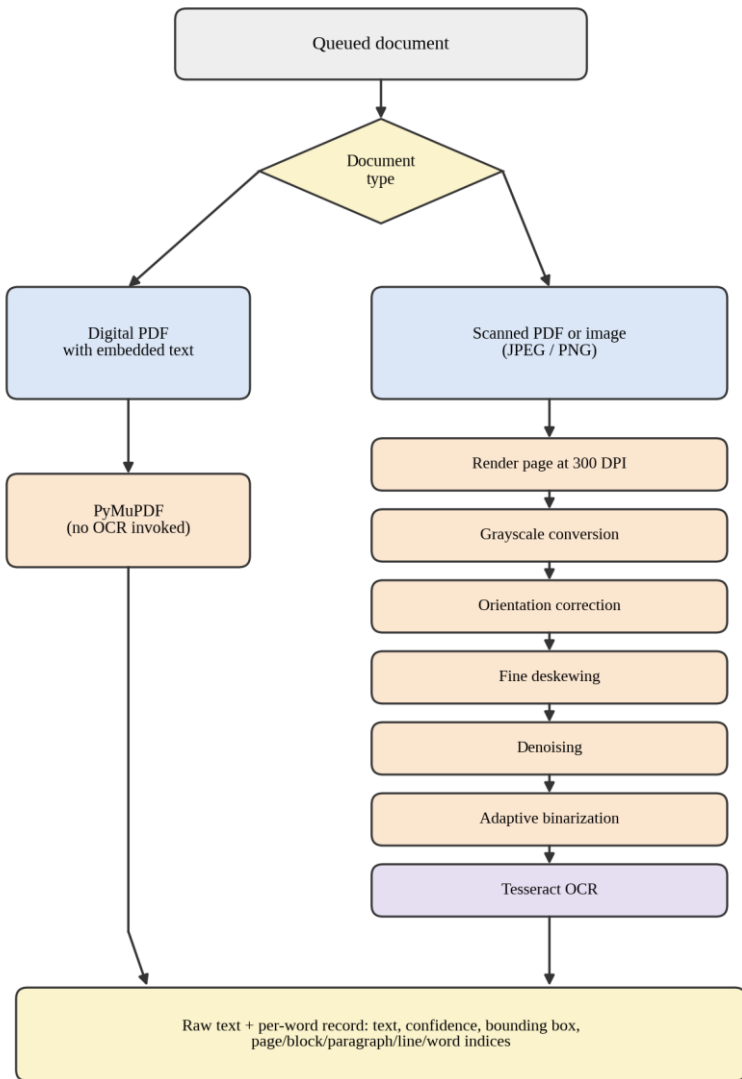


Figure 4.1: *The dual extraction path of module M2*

4.2.1 NATIVE-TEXT PDF PATH

Digitally generated PDFs that already contain embedded text are handled directly through PyMuPDF, which extracts that text without invoking OCR at all. The reported confidence for such documents is 1.0, since no visual recognition is involved.

4.2.2 SCANNED AND IMAGE PATH

Scanned PDFs and image uploads (JPEG, PNG) take the second route. Each page is rendered at 300 DPI and passed through an image-preprocessing sequence before Tesseract performs the character recognition. The sequence consists of grayscale conversion, coarse orientation correction, fine deskewing, denoising and adaptive binarization. Its qualitative effect is to convert a photographed or scanned page into the more uniform, higher-contrast input on which Tesseract performs best.

4.2.3 LAYOUT PRESERVATION

What happens after recognition is as important as the recognition itself. Instead of keeping a flat block of text, the implementation preserves Tesseract's per-word output: recognized text, confidence, a bounding box, and the page, block, paragraph, line and word indices for every word on the page. These word coordinates are then used to reconstruct the document's final raw text, instead of taking the text as-is in Tesseract's default reading order. This reconstruction attempts to do better than concatenating recognition output from top to

bottom, and it has not been benchmarked against the default ordering on pages with multiple text regions. The layout structure takes the following form:

```
{"pages": 1, "page_dimensions": [...],  
"page_confidences": [...], "words": [...]}
```

with each word entry recorded as:

```
{"page": 1, "text": "Medical",  
"confidence": 0.87,  
"bbox": {"x": 120, "y": 185,  
"width": 72, "height": 21}}
```

This full structure, comprising raw text, per-document confidence, page count, the OCR engine identifier and processing time, is persisted alongside the document and exposed through a dedicated text-retrieval endpoint. Both the recognized text and its spatial provenance therefore remain available to downstream stages.

4.2.4 LIMITATION: NO TABLE-STRUCTURE RECOGNITION

M2 does not perform table-structure recognition. Reading order across a page is reconstructed from word coordinates, but genuine table and cell structure, of the kind found in dense laboratory report grids, is not recovered as a table. It is recovered as positioned text.

4.3 PRIVACY DE-IDENTIFICATION (M3)

The text produced by M2, whether from native PDF parsing or OCR, is run through a de-identification layer (module M3) before any downstream processing touches it

(Figure 4.2). The layer is based on Microsoft Presidio's analyzer, augmented with custom recognizers for identifiers common in a healthcare setting that Presidio does not provide out of the box. Table 4.2 lists the recognizers used.

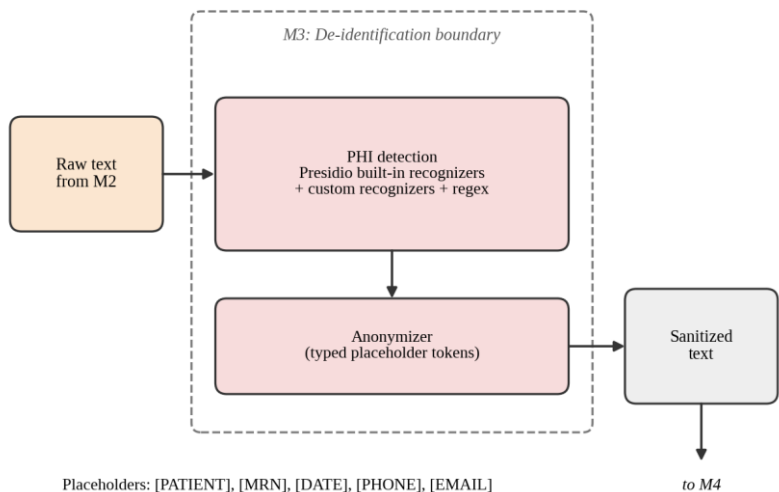


Figure 4.2: Privacy de-identification boundary

Table 4.2: Identifier recognizers used by the privacy layer

Source	Identifiers covered
Presidio built-in	Phone numbers, email addresses, physical addresses, dates of birth, IP addresses, account numbers
Custom (healthcare)	Patient name, medical record number (MRN), unique health identifier (UHID), patient ID, insurance or policy numbers

Detected identifiers are replaced with typed placeholder tokens such as [PATIENT], [MRN], [DATE], [PHONE] and [EMAIL] and are not deleted outright. This preserves the

surrounding sentence structure for the extraction stage that follows.

The privacy layer is intended to reduce exposure of identifiable information before downstream clinical processing. No claim is made that all protected health information has been removed. Automated de-identification is known to produce false negatives (identifiers that go undetected) and false positives (non-identifying text that is redacted unnecessarily), and this implementation is no exception. Section 5.6 reports what the layer actually did on the evaluation set, as an operational redaction count and not a measured precision or recall figure.

4.4 MEDICAL INFORMATION EXTRACTION (M4)

The extraction stage consumes the sanitized output of the privacy layer and produces the pipeline's clinical value (Figure 4.3). Instead of relying on a single model, M4 uses two complementary approaches whose results are merged.

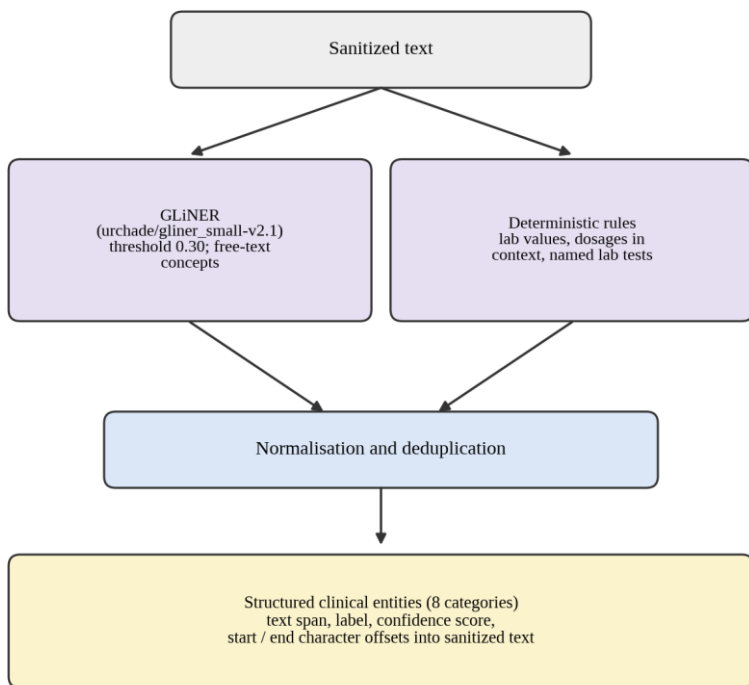


Figure 4.3: *Hybrid extraction pipeline of module M4*

4.4.1 ZERO-SHOT NER WITH GLiNER

A zero-shot clinical NER model, GLiNER (the `urchade/gliner_small-v2.1` checkpoint in the current configuration), detects mentions of eight clinical categories: Disease, Medication, Dosage, Symptom, Procedure, Anatomical Structure, Lab Test and Lab Value. The detection threshold is 0.30. Because the label set is supplied at inference time, no retraining is needed.

4.4.2 DETERMINISTIC RULE-BASED EXTRACTION

Deterministic rules extract the value types that are most likely to follow consistent patterns: laboratory values, dosages in context and named laboratory tests. For these categories pattern matching is likely to be more accurate and easier to audit than a learned model's output. See Table 4.3.

Table 4.3: *Clinical entity categories handled by M4*

Category	Example from the evaluation	Principal mechanism
Lab Value	192 mg/dL	Deterministic rules
Lab Test	eGFR	Deterministic rules
Dosage	500 mg	Deterministic rules
Medication	Metformin	GLiNER
Disease	diabetes	GLiNER
Procedure	Wide local excision, Tonometry	GLiNER
Symptom	Free-text findings	GLiNER
Anatomical Structure	Free-text findings	GLiNER

4.4.3 MERGING AND PROVENANCE

Results from both mechanisms are merged by a normalization step and a deduplication step before being stored as final structured entities. Each stored entity retains its extracted text span, its label, a confidence score, and character start and end offsets into the sanitized source text, and not merely the extracted value.

These offsets would enable an entity such as a medication name to be traced back to the specific position in the text it was extracted from. In conjunction with the OCR bounding-box information retained in M2, they would also allow a downstream stage to resolve that entity into a specific region of the original document image. Only what is implemented is claimed: entities retain character offsets into the sanitized text, and M2 retains the spatial OCR metadata needed for source-region resolution, but full resolution back to a highlighted document region is not evaluated in this work.

4.5 PERSISTENCE AND API LAYER

Extracted text, OCR layout and entities are all stored in a relational database, with layout and entities held as JSON/JSONB. The data is exposed through a set of versioned REST endpoints for uploading documents, retrieving text and retrieving entities. Keeping the full M2 structure (raw text, per-document confidence, page count, engine identifier and processing time) behind a dedicated text-retrieval endpoint is what lets later stages and external consumers work with spatial provenance and not just plain text.

4.6 USER INTERFACE

The frontend client through which users submit documents and follow their processing is a web application. Its main screens are described below; Figures 4.4 to 4.7 show them as implemented.

4.6.1 REGISTRATION AND LOGIN

New users create an account by entering their full name, work email and a password of at least eight characters, and by selecting a clinical role from a drop-down list (Figure 4.4). Registered users sign in with their work email and password (Figure 4.5).

The screenshot shows the HealthParse registration interface. On the left, a teal sidebar contains the HealthParse logo and the heading "Create Your HealthParse Account". Below the heading, it states: "Join your healthcare organization on the intelligent document processing platform designed for clinical workflows." Three bullet points with checkmarks list features: "Role-tailored clinical access", "High-capacity document processing", and "Instant validation & audit readiness". At the bottom of the sidebar is the text "HealthParse Clinical Document Intelligence Platform © 2026". On the right, the "Create account" section has the subtext "Register to begin processing clinical documents". It includes input fields for "Full Name" (with "Dr. Jane Okafor" as an example), "Work Email" (with "you@hospital.org" as an example), and "Password" (with a note "At least 8 characters"). Below these is a "Clinical Role" dropdown menu with "Doctor" selected. A teal "Create Account" button is at the bottom of the form, with a link "Already registered? Sign in" below it.

Figure 4.4: Registration page

The screenshot shows the HealthParse login interface. On the left, a teal sidebar contains the HealthParse logo and the heading "Automated Healthcare Document Workspace". Below the heading, it states: "Secure, centralized document ingestion and tracking built for healthcare professionals, clinical staff, and medical records administrators." Three bullet points with checkmarks list features: "Streamlined lab reports, prescriptions & records", "HIPAA-aligned security & role controls", and "Real-time processing status tracking". At the bottom of the sidebar is the text "HealthParse Clinical Document Intelligence Platform © 2026". On the right, the "Sign in" section has the subtext "Access your clinical document portal". It includes input fields for "Work Email" (with "james@dhospital.org" as an example) and "Password" (with "*****" as an example). A teal "Sign in to HealthParse" button is at the bottom of the form, with a link "Need access? Create an account" below it.

Figure 4.5: Login page

4.6.2 DASHBOARD

After sign-in the dashboard gives an overview of intake activity (Figure 4.6). Summary cards show the total number of documents and how many are processing, completed or failed. A Recent Documents table lists each document with its name, type, upload time and processing status (for example Pending or Completed), which reflects the asynchronous processing described in Section 4.1. A button on the dashboard opens the upload page, and the side navigation links the Dashboard, Documents and Upload Document views.

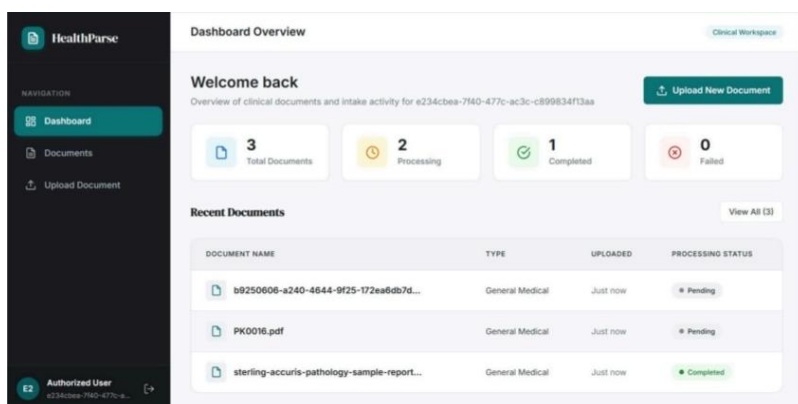


Figure 4.6: Dashboard overview

4.6.3 DOCUMENT UPLOAD

The upload page, titled Document Ingestion Workspace, accepts lab reports, discharge summaries, prescriptions and medical images (Figure 4.7). A document can be dragged onto the drop area or chosen by browsing. The page lists PDF, JPG, JPEG and PNG as the accepted types and shows a maximum

file size of 20 MB. Submitting the file triggers the ingestion stage (M1).

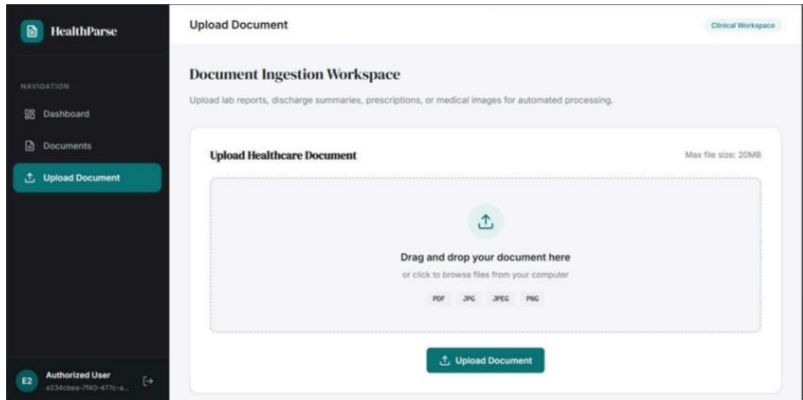


Figure 4.7: *Document upload page*

4.7 MODULES OUTSIDE THE SCOPE OF THIS REPORT

The wider platform includes additional downstream capabilities, most notably document-type classification. They are not evaluated, diagrammed or claimed as contributions here. This is stated explicitly so that they are not assumed to be part of the work reported.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The pipeline is implemented as a FastAPI backend in Python. Native-PDF text extraction is performed with PyMuPDF, and scanned documents and images are read with Tesseract OCR behind an OpenCV-based preprocessing chain that covers grayscale conversion, orientation detection and correction, deskewing, denoising and adaptive binarization. The privacy de-identification component is built on Microsoft Presidio's analyzer and anonymizer with custom healthcare-specific recognizers on top. For medical information extraction, GLiNER (urchade/gliner_small-v2.1) is used together with a custom deterministic rule-based system for lab values, dosages and lab tests.

5.2 EVALUATION SETUP

Consistent with the scope of this report, the evaluation examines the actual execution behaviour of the implemented M1 to M4 pipeline on a fixed six-document set, and not a model-performance benchmark against labelled ground truth. The set intentionally mixes format types. It comprises two native-text PDFs (a 13-page and a 19-page clinical report) and four scanned or image documents (three JPEGs and one scanned PDF) processed through the Tesseract OCR path.

The evaluation was conducted on a 12th Gen Intel Core i5-1235U CPU with 8 GB RAM and integrated Intel UHD Graphics, in a CPU-only configuration with no discrete GPU (Table 3.2). The software environment used Python 3.10.0, Tesseract OCR 5.5.3.20260724 (Leptonica 1.87.0) and the GLiNER model `urchade/gliner_small-v2.1`. The same environment and processing configuration were used for all six documents.

5.3 FUNCTIONAL VERIFICATION

Before reporting pipeline output, correct end-to-end operation of each stage was verified. The implementation's test suite exercises document ingestion and validation, the native-PDF and OCR extraction pathways, privacy de-identification, and the extraction and merging logic in M4. It is executed as part of the project's standard verification procedure ahead of any evaluation results. Functional verification, which is confirmation that each stage behaves as designed on its test inputs, is a separate matter from the output-quality results that follow.

5.4 AGGREGATE PIPELINE EXECUTION RESULTS

All six documents were processed successfully with no pipeline failures and zero rows in the error record, giving a 100% execution success rate for this set. Table 5.1 gives the per-document breakdown, and Table 5.2 summarizes the timing. In Table 5.1 the M4 column (marked with an asterisk)

is obtained as total time minus M2 and privacy time, and the 13-page native PDF value is the logged 10301.76 ms. Figure 5.1 shows how the time divides between stages.

Table 5.1: *Per-document execution results across the six-document evaluation set*

Document (type)	Pg.	Engine	M2 (ms)	M3 (ms)	M4* (ms)	Total (ms)	Ent.
PK0016 (PDF)	13	Native	143	2059	10302	12503	68
Sterling path. rpt. (PDF)	19	Native	204	3135	709	4048	33
Tmpl. 1 sample 1 (PDF)	1	Tesseract	6126	218	641	6985	15
Tmpl. 2 sample 5 (JPG)	1	Tesseract	3343	167	582	4092	10
Tmpl. 4 sample 1 (JPG)	1	Tesseract	3375	201	585	4161	23
Tmpl. 7 sample 7 (JPG)	1	Tesseract	2750	113	319	3182	15
Mean	-	-	2657	982	2189	5829	27.3

Table 5.2: *Summary of processing time and success rate*

Measure	Value
Documents processed successfully	6 of 6 (100%)
Mean total processing time	5828.55 ms
Median total processing time	4126.48 ms
Fastest document	3182.28 ms
Slowest document (13-page native PDF)	12502.96 ms
Mean M2 time (preprocessing / OCR)	2656.87 ms
Mean M3 (privacy) processing time	982.18 ms
Mean M4 extraction time	2189.48 ms

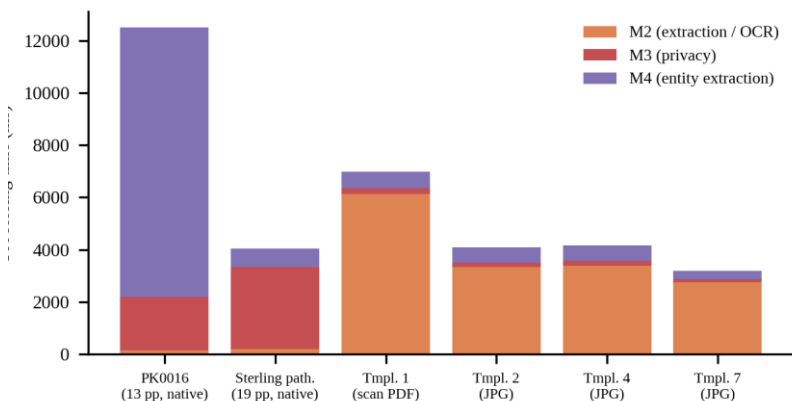


Figure 5.1: *Per-document processing time by stage*

The slowest document was the 13-page native PDF, driven by an M4 extraction time of 10301.76 ms against a much larger volume of embedded laboratory data. For the scanned single-page documents, M2 (preprocessing and OCR) dominated, whereas for the native PDFs, which need no OCR, M2 took only a few hundred milliseconds. These figures are reported as an operational demonstration of the pipeline's execution characteristics on this six-document set and not as a claim about throughput on an arbitrary document population.

5.5 OCR AND PREPROCESSING CHARACTERISTICS

The two native-text PDFs were parsed with a reported PyMuPDF confidence of 1.0000, since no visual recognition

was required. The four scanned or image documents processed through Tesseract reported per-document confidence between 0.9301 and 0.9478, consistent across formats despite differing image quality and content density.

This confidence reflects Tesseract's own certainty about its reading and is not a measured accuracy against ground truth. It is best used as a signal for flagging text that deserves closer review and not as a substitute for such a measurement. The preprocessing sequence, from grayscale conversion to adaptive binarization, is applied uniformly ahead of Tesseract on the scanned and image path.

5.6 PRIVACY DE-IDENTIFICATION (M3) RESULTS

Across the six documents the privacy layer recorded 484 redactions spanning eleven identifier categories: DATE_TIME, PERSON, PATIENT_NAME, PATIENT_ID, ACCOUNT_NUMBER, LOCATION, IP_NUMBER, PHONE_NUMBER, ADDRESS, DATE_OF_BIRTH and MRN.

Redaction volume scaled with document length and identifier density rather than with format. The 19-page pathology report produced the most redactions (247, dominated by 98 PERSON and 107 DATE_TIME detections), while the single-page scanned documents each produced between 21 and 34 redactions.

These counts are the layer's recorded operational output, which means what it detected and replaced. They are not a

measured precision or recall figure, because no labelled identifier ground truth evaluation has been run against this set.

5.7 CLINICAL ENTITY EXTRACTION RESULTS

M4 identified 164 structured clinical entities across the six documents, populating all eight supported entity labels. Table 5.3 shows the distribution and Figure 5.2 shows it graphically.

Table 5.3: *Extracted entity counts by label across the six-document set*

Entity label	Count	Documents present in
Lab Value	56	4 / 6
Lab Test	51	3 / 6
Symptom	18	4 / 6
Anatomical Structure	16	4 / 6
Procedure	14	6 / 6
Disease	5	2 / 6
Medication	3	3 / 6
Dosage	1	1 / 6
Total	164	-

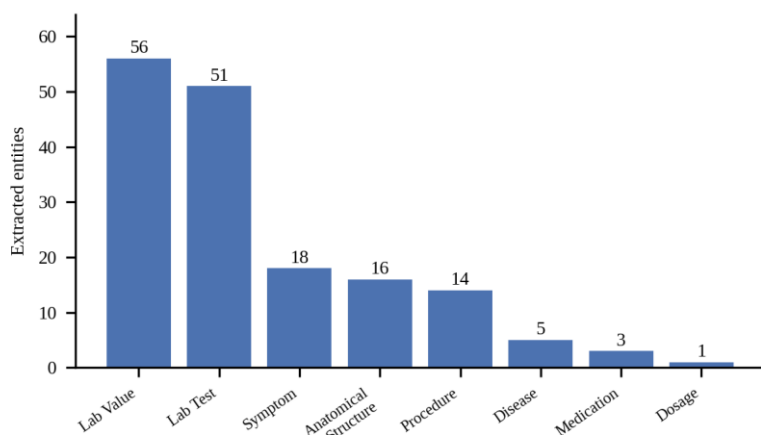


Figure 5.2: *Distribution of the 164 extracted entities by label*

Lab Value (56) and Lab Test (51) entities dominated the two native-PDF laboratory reports, where deterministic pattern rules apply most readily to the structured numeric-plus-unit format those documents use (for example "192 mg/dL" and "eGFR"). Symptom, Anatomical Structure, Disease and Procedure entities, the categories that depend most on the zero-shot GLiNER model, were concentrated in the four scanned clinical-note and radiology-style documents, where findings are expressed in free text and not in tabular values.

Procedure was the only label present in all six documents, in contexts ranging from a laboratory procedure ("Microscopy") to surgical and diagnostic procedures ("Wide local excision", "Slit-lamp Examination", "Tonometry"). Every entity record retained a label, a confidence score and character

start and end offsets into the sanitized text, consistent with the provenance design in Section 4.4. Entity confidences ranged from 0.30, the GLiNER detection threshold, for ambiguous free-text spans, up to 0.95 for deterministically matched lab values.

5.8 QUALITATIVE END-TO-END EXAMPLE

The following illustrative example was constructed to show stage-to-stage behaviour on a single sentence. It is not one of the six measured evaluation records. Consider a sentence of the kind M2 recovers from a scanned document:

Patient John Doe, DOB 04/12/1985, MRN 88213,
was prescribed Metformin 500 mg for diabetes.

The privacy layer reduces this to the following, leaving the clinical content untouched:

Patient [PATIENT], DOB [DATE], MRN [MRN],
was prescribed Metformin 500 mg for diabetes.

M4 then extracts three structured entities from the sanitized sentence:

```
{ "text": "Metformin", "label": "MEDICATION",  
  "start_char": 61, "end_char": 70}  
{ "text": "500 mg", "label": "DOSAGE",  
  "start_char": 71, "end_char": 77}  
{ "text": "diabetes", "label": "DISEASE",  
  "start_char": 82, "end_char": 90}
```

Each of these entities is present in, and only in, the sanitized sentence, so no clinical value appears in the output that was not in the input text. MEDICATION and DISEASE come from the zero-shot model, while DOSAGE suits the

pattern-based rule path. This is representative of how the two mechanisms divide the work across the eight categories in the evaluation set as a whole.

5.9 DISCUSSION

The results show that heterogeneous healthcare documents, long native-text PDFs and short scanned images alike, can be processed through a single pipeline into structured, de-identified and traceable clinical output, with zero recorded failures across all six evaluation documents. What makes the pipeline interesting as a system is not the peak accuracy of any one stage but that the stages compose well. OCR output keeps the layout metadata that downstream traceability needs, privacy processing performed 484 redactions while leaving the clinical text available for extraction, and extraction retained 164 entities with character offsets so that the output is auditable and not opaque.

The split between entity categories and document types, with lab values and lab tests concentrated in the two long structured PDF reports and symptoms and anatomical findings concentrated in the four shorter free-text scans, is indicative of the intended division between deterministic extraction of structured patterns and model-based extraction of free-text concepts. This is a descriptive observation, not a controlled experiment on the behaviour of the two mechanisms.

5.10 WHAT THIS EVALUATION DOES NOT ESTABLISH

- It does not provide manually annotated ground truth, so there is no entity-level precision, recall or F1. The entity and redaction counts are the pipeline's own operational output.
- It does not provide an independently validated privacy recall or precision figure. The 484 redactions are recorded detections and anonymization operations, not a confirmed true-positive count.
- Successful execution on six documents is not evidence of large-scale production scalability or clinical reliability. The set, two long native-PDF lab reports and four short scanned clinical notes, is small and was not built as a statistically representative sample.
- The reconstructed reading order from word coordinates has not been benchmarked against Tesseract's default ordering on pages with multiple text regions.
- The current M4 configuration does not use explicit long-document chunking, so documents that exceed the model's sequence length may be truncated during GLiNER inference, while deterministic rules still apply to the entire sanitized text.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This report described **HealthParse**'s ingestion-through-extraction pipeline: document validation and asynchronous processing; dual native-PDF and OCR text extraction with layout fidelity and confidence metadata; Presidio-based privacy de-identification; and hybrid GLiNER-plus-rules clinical entity extraction with character-offset provenance.

In an end-to-end evaluation on a six-document mixed-format corpus, 100% of documents completed execution, the mean processing time was 5.83 seconds per document, 164 structured clinical entities were extracted across eight categories, and 484 privacy redactions were recorded across eleven identifier types. The contribution is the integrated, implemented pipeline and the concrete execution evidence collected against it. It is not a claim of state-of-the-art accuracy in any single stage, and not a claim that identifying information is guaranteed to be removed.

6.2 FUTURE WORK

- A labelled-ground-truth accuracy evaluation of both de-identification (precision and recall of identifier detection) and clinical entity extraction (entity-level precision, recall and F1).

- Expansion of the evaluation set beyond the present six documents to a larger and more representative sample.
- Extending character-offset traceability to full source-region resolution, so that an entity can be highlighted on the original document image using the OCR layout metadata already preserved in M2.
- Benchmarking the coordinate-based reading-order reconstruction against Tesseract's default ordering on multi-region pages.
- Adding explicit long-document chunking to the M4 stage to avoid truncation during GLiNER inference.
- Table-structure recognition for dense laboratory grids, and, separately from this report, the platform's downstream document-classification capability.

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HealthParse: A Modular Pipeline for Privacy-Aware Extraction of Structured Clinical Information from Heterogeneous Healthcare Documents

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Abstract—A hospital intake desk can receive a cleaned typeset lab PDF, a faxed discharge summary, and a phone photo of a handwritten prescription in the same hour, and all must become useful clinical data rather than just be stored as incomprehensible files. HealthParse is a document processing pipeline designed to tackle such mixed document types, carrying a healthcare document through four stages: ingestion and validation, format-aware text extraction (native-text PDFs through direct parsing, scanned or photographed documents through Tesseract OCR and image preprocessing), privacy-aware de-identification of the recovered text, and hybrid clinical information extraction (a zero-shot named-entity model plus deterministic rules for values like dosages and lab results). Instead of showing only the conceptual architecture of this flow, this paper reveals what each stage actually produces (the data structures, confidence scores, and character-level provenance it retains) and reports an end-to-end evaluation on a six-document corpus mixing native-text PDFs and scanned images: all six documents were processed successfully, the pipeline produced 164 structured clinical entities across eight categories, and the privacy layer recorded 484 identifier redactions across eleven identifier types, at a mean end-to-end processing time of 5.83 seconds per document. These results are framed as operational execution results rather than an accuracy benchmark against labeled ground truth, and we are always explicit about that distinction. This paper is scoped deliberately to the ingestion-through-extraction chain and not to the full table-structure recognition or guaranteed removal of all identifying information or any kind of clinical decision support.

Keywords—Healthcare Document Processing, Optical Character Recognition, Clinical Named Entity Recognition, Medical Information Extraction, Privacy-Preserving De-identification, Document Intelligence

I. INTRODUCTION

Walk into most healthcare administrative offices and you will find records that don't share a common format [1], [2]. A lab returns a digitally generated PDF report containing neatly embedded text. A referring clinic faxes an attached discharge summary, scanned in as a low-quality, wavy image. A patient photographs a handwritten prescription on their phone, then uploads the picture through a portal. All three are essentially the same kind of information, but only the first is machine-parsable directly. The last two (and many similar ones) require an intermediary step of optical character recognition before any downstream system can do anything with them, which is far from a perfect process: its accuracy is

highly dependent on scan quality, lighting conditions, document skew, and other factors [3], [4].

This is a nontrivial limitation. Once a document is converted to text, it typically still contains identifying patient information, which must be handled appropriately before any of the content can be used for its intended purpose. And reliably performing all this processing, for documents in a variety of formats the system may not have been specifically trained on, is a significant engineering task in and of itself, distinct from and arguably more challenging than training an extraction model capable of doing so.

This paper describes the HealthParse document processing pipeline, designed to handle precisely this task. Given a healthcare-related document in one of several common formats, the pipeline ingests and validates the document, extracts any embedded machine-readable text (using native PDF tools when available) or performs OCR (with Tesseract [5], following an image preprocessing sequence [6], [7]) to obtain text from the document image, de-identifies the extracted text, then extracts structured clinical information from the resulting text using a combination of zero-shot named-entity recognition and deterministic pattern extraction (for example, for lab results or dosages).

This paper is scoped to focus specifically on this chain: ingestion and OCR/preprocessing, privacy-preserving de-identification of the extracted text, and extraction of the relevant clinical information. The pipeline has additional capabilities not described in this paper, most notably document classification, that fall outside the scope of this discussion, which we explicitly revisit in Section VIII. Within the scope of this paper, its contribution is:

- Description of the implemented pipeline at the level of concrete data structures/API responses, covering processing decisions that go beyond a conceptual architecture diagram
- Detailed discussion of the OCR/preprocessing stage's dual processing path for native-text PDFs vs scanned/image documents, covering what layout information gets preserved (rather than discarded) during text extraction
- Description of the privacy layer's capabilities and limitations: what sort of information it was designed to reduce exposure of, and an acknowledgment that automated de-identification of this sort can both over-redact and under-redact

- Description of the hybrid clinical extraction stage combining zero-shot NER with deterministic rules across eight clinical entity categories, including discussion of the character-offset provenance tracking that links each extracted entity to a location in the original text
- End-to-end system evaluation on a six-document test set containing a mix of native PDFs and scanned images, reporting overall execution success rate, per-stage timing breakdown, entity extraction counts by label, and privacy redactions applied, along with discussion of what lies outside this system’s scope of evaluation

II. RELATED WORK

A. OCR for Healthcare Documents

OCR has been studied for general digitization of documents [8]–[11], and for specific domains (invoices [12]–[14], historical archives [15], [16]), but relatively little work has focused on healthcare-specific benchmarking [17], [18]. The literature agrees that OCR performance is not a static value, but varies with the quality of documents (their layout complexity) and the degree of degradation in the image (the scan), and that errors at this stage can affect subsequent text processing steps [3]–[5]. We do not attach specific accuracy figures to this general statement, since the exact values reported in any one study are specific to that study’s data and evaluation setup and cannot be directly applied to a different document population.

B. Clinical Named Entity Recognition

The extraction of structured clinical entities from free text has traditionally been performed using supervised models trained on annotated corpora from the relevant clinical domain [19], [20]. Recent advances in zero-shot and few-shot span-identification models like GLiNER [21] enable the specification of a flexible label set at inference time without requiring retraining, at the cost of some raw accuracy in favor of being able to adapt without fresh labeled data to new entity categories. Deterministic, rule-based extraction remains common practice alongside learned models for structured value types —lab measurements, dosages, dates—where a pattern-based approach can often yield better results and be more interpretable than a learned model would be for that narrow category.

C. Privacy-Preserving De-identification

De-identification of clinical text is often approached as a mixture of rule-based recognizers for identifiers that take a predictable form (phone numbers, dates, structured ID numbers) and statistical or learned recognizers for identifiers that appear in unstructured free text, such as patient names [22]. Toolkits such as Microsoft Presidio [23] realize this mixture as a configurable pipeline of recognizers and anonymization operators and are commonly enhanced with custom recognizers for identifiers common to a given domain, such as medical record numbers in a healthcare setting. It is well understood in this literature that there is no automated de-identification approach that provides a guarantee of complete removal; false negatives (missed identifiers) and false positives (over-redaction of non-identifying text) are both known failure modes, rather than edge cases.

D. Positioning of This Work

Much of the existing literature in this space works on one stage (say, an OCR engine comparison) in isolation, with far less said about how these stages actually integrate into a single running pipeline and the implementation choices that this integration entails (how to represent OCR layout metadata once extraction moves to a downstream stage, how to pass de-identified text forward while retaining document structure, how to retain a traceable link from extracted entities back to their source). This paper’s contribution is at the level of integration, articulated concretely and demonstrated on a mixed-format document evaluation set rather than in the abstract.

III. SYSTEM ARCHITECTURE

HealthParse is implemented as a layered application with a frontend client submitting documents to a FastAPI backend that orchestrates downstream processes related to document ingestion, OCR/preprocessing, privacy de-identification, and clinical information extraction, with results persisted to a relational database along with the originally uploaded files. Fig. 1 provides an overview of the processing pipeline from the perspective of a single document processed by the application, from ingestion, through document-type-specific OCR (M2), through privacy-aware de-identification, to the structured clinical entities that the pipeline ultimately produces (M4). Sections IV and V discuss each of these stages in turn.



Fig. 1. End-to-end HealthParse pipeline: document ingestion (M1), document processing/OCR (M2), privacy-aware de-identification, and hybrid clinical information extraction (M4), producing structured clinical information with entity labels, confidence scores, and source offsets.

IV. METHODOLOGY

A. Document Ingestion (M1)

Documents are uploaded via a REST endpoint, which accepts PDF, JPEG, and PNG file types; each uploaded file is validated against the MIME type that was declared, and then a UUID is generated and associated with the uploading user (and an optional patient identifier), before the file is written to storage. A document record is created with a Processing status, and the remaining pipeline stages are scheduled in an asynchronous fashion, instead of being executed inline with the upload request. This design decouples document ingestion latency from the time required for OCR and subsequent stages to complete. We refer to this stage as ingestion and registration, rather than upload, because this behavior affects how the rest of the pipeline is composed and reasoned about.

B. Preprocessing and OCR (M2)

Once a document is queued for processing, the system decides between two extraction paths depending on document type. Digitally generated PDFs that already contain embedded text are handled directly through PyMuPDF, which extracts that text without invoking OCR at all—there is nothing to recognize visually when the text is already encoded in the file. Scanned PDFs and image uploads (JPEG, PNG) take a

different route: each page is rendered at 300 DPI and passed through an image-preprocessing sequence [6], [7]—grayscale conversion, coarse orientation correction, fine deskewing, denoising, and adaptive binarization—before Tesseract [5] performs the actual character recognition.

What happens after we recognize the text is just as important as the recognition itself. Instead of keeping a flat block of recognized text, the implementation preserves Tesseract’s per-word output: recognized text, confidence, a bounding box, and page/block/paragraph/line/word indices for every word on the page. We then use these word coordinates to reconstruct the document’s final raw text, instead of taking it as-is from Tesseract’s default reading order. This approach attempts to do better than just concatenating recognition output from top to bottom, and has not been benchmarked against the default ordering on pages with multiple text regions. The layout structure will take the form:

```
{
  "pages": 1,
  "page_dimensions": [...],
  "page_confidences": [...],
  "words": [...]
}
```

with each word entry recorded as:

```
{
  "page": 1,
  "text": "Medical",
  "confidence": 0.87,
  "bbox": {
    "x": 120,
    "y": 185,
    "width": 72,
    "height": 21
  }
}
```

This full structure—raw text, per-document confidence, page count, the OCR engine identifier, and processing time—is persisted alongside the document and exposed through a dedicated text-retrieval endpoint, so that both the recognized text and its spatial provenance remain available to downstream stages rather than being discarded once extraction completes. One limitation is worth stating plainly here rather than leaving implicit: M2 does not perform table-structure recognition. Reading order across a page is reconstructed from word coordinates, but genuine table and cell structure—the kind found in dense laboratory report grids—is not recovered as a table; it is recovered as positioned text.

C. Privacy Processing

The text produced by M2, whether from native PDF parsing or OCR, is run through a de-identification layer before any downstream processing touches it (Fig. 2). This layer is based on Microsoft Presidio’s analyzer [23], augmented with custom recognizers for identifiers common to a healthcare setting that Presidio doesn’t have out of the box—patient name, medical record number (MRN), unique health identifier (UHID), patient ID, and insurance or policy numbers—used in conjunction with Presidio’s built-in recognizers for phone numbers, email addresses, physical addresses, dates of birth, IP addresses, and account numbers. Detected identifiers are replaced with typed placeholder tokens such as [PATIENT], [MRN], [DATE], [PHONE], and [EMAIL] rather than being deleted outright, preserving the surrounding sentence structure for the extraction stage which comes next.

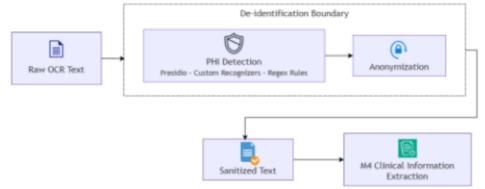


Fig. 2. Privacy de-identification boundary: PHI detection (Presidio, custom recognizers, and regex rules) followed by anonymization, producing sanitized text that is handed forward to M4 clinical information extraction.

We’re being careful about what we say about this stage. The privacy layer is intended to reduce exposure of identifiable information prior to downstream clinical processing; we make no claim here that all protected health information has been removed. Automated de-identification of this sort is known to produce false negatives—identifiers that go undetected—and false positives—non-identifying text that gets redacted unnecessarily—in general, and this implementation is no exception. Section VI-C and VI-D report what this layer actually did on our six-document evaluation set, as an operational redaction count rather than a measured precision/recall figure against a labeled identifier ground truth (which has as yet not been run).

D. Medical Information Extraction (M4)

The extraction stage consumes the sanitized output of the privacy layer and produces the pipeline’s clinical value (Fig. 3). Instead of using a single model, M4 uses two complementary extraction approaches. First, a zero-shot clinical NER model (GLiNER [21])—urchade/gliner_small-v2.1 checkpoint in the current configuration—is used to detect mentions of eight clinical categories: Disease, Medication, Dosage, Symptom, Procedure, Anatomical Structure, Lab Test, and Lab Value with a detection threshold of 0.30. Second, deterministic rule-based extraction is used to extract value types that are more likely to have consistent patterns than the free text—laboratory values, dosages in context, and named laboratory tests—where pattern matching is likely to be more accurate and easier to audit than a learned model’s output for the same category.

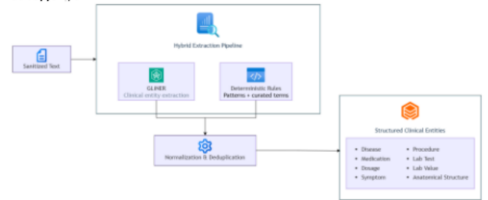


Fig. 3. M4 hybrid extraction pipeline: sanitized text is processed in parallel by GLiNER (zero-shot clinical entity extraction) and a deterministic rule engine, with results merged through normalization and deduplication into structured clinical entities across eight categories.

Results from both mechanisms are merged by a normalization step and a deduplication step before being stored as final structured entities. Each such stored entity retains its extracted text span, its label, a confidence score, and character start/end offsets into the sanitized source text (not merely the extracted value). It is these character offsets that

would enable an entity such as an extracted medication name to be traced back to the specific position in the text from which it was extracted, which in turn (in conjunction with the OCR bounding-box information retained in M2) would allow a downstream stage to resolve that entity into a specific region of the original document image. We are careful to note only what is implemented at this stage: extracted entities retain character offsets into the sanitized text, and M2 retains the spatial OCR metadata necessary to enable source-region resolution, but full resolution back to a highlighted document region is not evaluated in this paper.

V. IMPLEMENTATION

The pipeline outlined above has been implemented as a FastAPI backend in Python. Native-PDF text extraction is performed using PyMuPDF, while for scanned documents and images, Tesseract OCR is used with an accompanying OpenCV-based preprocessing chain that includes grayscale conversion, orientation detection and correction, deskewing, denoising, and adaptive binarization. The privacy de-identification component is based on Microsoft Presidio’s analyzer and anonymizer with custom healthcare-specific recognizers added on top. For medical information extraction, we use GLiNER (urchade/gliner_small-v2.1) in conjunction with a custom deterministic rule-based system for lab values, dosages, and lab tests. Extracted text, OCR layout, and entities are all stored in a relational database, with layout and entities being stored as JSON/JSONB, and exposed through a set of versioned REST endpoints for uploading documents, retrieving text, and retrieving entities.

VI. SYSTEM DEMONSTRATION AND EVALUATION

Consistent with this paper’s scope, we evaluate the implemented M1–M4 pipeline’s actual execution behavior on a fixed six-document set, rather than presenting a model-performance benchmark against labeled ground truth as the central result. The evaluation set intentionally mixes format types: two native-text PDFs (a 13-page and a 19-page clinical report) and four scanned/image documents (three JPEGs and one scanned PDF) processed through the Tesseract OCR path. This section covers functional verification, aggregate pipeline execution results, OCR/preprocessing characteristics, privacy de-identification results, clinical entity extraction results, and a single worked qualitative example. Where a quantity was not captured by this evaluation run, we say so directly in Section VI-G rather than filling in a plausible-sounding number.

Evaluation Environment: The evaluation was conducted on a 12th Gen Intel Core i5-1235U CPU with 8 GB RAM and integrated Intel UHD Graphics (CPU-only configuration; no discrete GPU was used). The software environment used Python 3.10.0, Tesseract OCR 5.5.3.20260724 (Leptonica 1.87.0), and the GLiNER model urchade/gliner_small-v2.1. The same environment and processing configuration were used for all six documents.

A. Functional Verification

Before reporting on pipeline output, we verify correct end-to-end operation of each stage. The implementation’s test suite exercises document ingestion and validation, the native-PDF and OCR extraction pathways, privacy de-identification, and the extraction and merging logic in M4, and is executed as part of the project’s standard verification

procedure ahead of any evaluation results reported in this section. We view functional verification —confirmation that each stage operates as designed on its test inputs —as a separate consideration from the output-quality results that follow.

B. Aggregate Pipeline Execution Results

Using the six document evaluation set, we processed all six documents successfully with no pipeline failures, logging zero rows in the error record, for 100% execution success rate for this set. Table I shows the complete per-document breakdown, including input format and page count, the OCR/extraction engine used and its confidence, per-stage processing time (M2 OCR/preprocessing, privacy de-identification, and M4 extraction), total processing time, and the number of structured entities produced.

TABLE I
PER-DOCUMENT EXECUTION RESULTS ACROSS THE SIX-DOCUMENT EVALUATION SET

Document (type)	Pg.	Engine	M2 (ms)	Priv. (ms)	Tot. (ms)	Ent.
PK0016 (PDF)	13	native	143	2059	12503	68
Sterling path. rpt. (PDF)	19	native	204	3135	4048	33
Tmpl. 1 sample 1 (PDF)	1	Tesseract	6126	218	6985	15
Tmpl. 2 sample 5 (JPG)	1	Tesseract	3343	167	4092	10
Tmpl. 4 sample 1 (JPG)	1	Tesseract	3375	201	4161	23
Tmpl. 7 sample 7 (JPG)	1	Tesseract	2750	113	3182	15
Mean	—	—	2657	982	5829	27.3
Median	—	—	—	—	4126	—

All six documents were marked SUCCESS, with a mean total processing time of 5828.55 ms and a median of 4126.48 ms; the fastest document completed in 3182.28 ms and the slowest—the 13-page native PDF, driven by its M4 extraction time of 10301.76 ms against a much larger volume of embedded lab data —completed in 12502.96 ms. Averaged across the set, M2 (preprocessing/OCR) accounted for a mean of 2656.87 ms, privacy processing for 982.18 ms, and M4 extraction for 2189.48 ms per document. We report these as an operational demonstration of the implemented pipeline’s execution characteristics on this six-document set, not as a claim about throughput on an arbitrary document population.

C. OCR and Preprocessing Characteristics

The two native-text PDFs were parsed with a reported PyMuPDF confidence of 1.0000, since no visual recognition was required. The four scanned/image documents processed through Tesseract reported per-document confidence between 0.9301 and 0.9478, consistent across formats despite differing image quality and content density. As emphasized in Section IV-B, this confidence score reflects Tesseract’s own certainty about its reading, not a measured accuracy against ground truth; it is a signal for flagging text worth closer review rather than a substitute for one. The preprocessing sequence —grayscale conversion through adaptive binarization —is applied uniformly ahead of Tesseract on the scanned/image path, and its qualitative effect is to convert a photographed or scanned page into the more uniform, higher-contrast input that Tesseract performs best on.

D. Privacy De-identification Results

Across the six documents, the privacy layer recorded 484 total redactions spanning eleven identifier categories:

DATE_TIME, PERSON, PATIENT_NAME, PATIENT_ID, ACCOUNT_NUMBER, LOCATION, IP_NUMBER, PHONE_NUMBER, ADDRESS, DATE_OF_BIRTH, and MRN. Redaction volume scaled with document length and identifier density rather than format: the 19-page pathology report produced the most redactions (247, dominated by 98 PERSON and 107 DATE_TIME detections), while the single-page scanned documents each produced between 21 and 34 redactions. As in Section IV-C, we report this as the layer’s recorded operational output —what it detected and replaced —rather than as a measured precision or recall figure, since no labeled identifier ground truth evaluation has been run against this set.

E. Clinical Information Extraction Results

M4 identified 164 structured clinical entities in the six documents, populating all eight supported entity labels. Table II shows the distribution; Fig. 4 shows the same distribution graphically. Lab value (56) and lab test (51) entities dominated the two native-PDF laboratory reports, where deterministic pattern rules are most applicable to the structured numeric+unit format those documents use (e.g., “192 mg/dL”, “eGFR”). Symptom, anatomical structure, disease, and procedure entities —the categories that most rely on the zero-shot GLINER model rather than pattern rules —were concentrated in the four scanned clinical-note and radiology-style documents, where clinical findings are expressed in free text rather than tabular values.

TABLE II

EXTRACTED ENTITY COUNTS BY LABEL ACROSS THE SIX-DOCUMENT SET

Entity label	Count	Docs. present in
Lab Value	56	4/6
Lab Test	51	3/6
Symptom	18	4/6
Anatomical Structure	16	4/6
Procedure	14	6/6
Disease	5	2/6
Medication	3	3/6
Dosage	1	1/6
Total	164	–

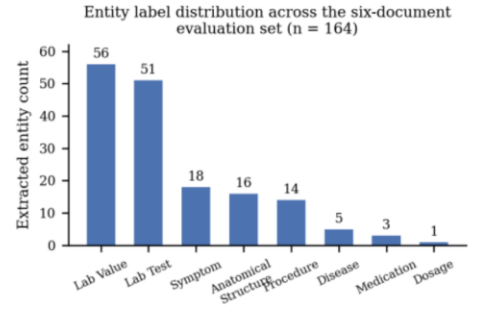


Fig. 4. Distribution of the 164 structured clinical entities extracted across the six-document evaluation set, by entity label.

Procedure was the only label which appeared in all six documents, used in contexts ranging from a microscopy laboratory procedure (“Microscopy”) to surgical and

diagnostic procedures (“Wide local excision”, “Slit-lamp Examination”, “Tonometry”), reflecting that some description of a procedure performed is common to lab reports, dermatology/pathology notes and clinical encounter notes alike. Every entity record retained a label, a confidence score, and character start/end offsets into the sanitized source text, consistent with the provenance design described in Section IV-D; individual entity confidences in this run ranged from as low as 0.30 (the GLINER detection threshold) for ambiguous free-text spans up to 0.95 for deterministically matched lab values.

F. Qualitative End-to-End Example

The following is an illustrative system-output example constructed to show the pipeline’s stage-to-stage behavior on a single sentence; it is not one of the six measured evaluation records reported in Sections VI-B through VI-E. Consider a representative sentence of the kind M2 recovers from a scanned document: “Patient John Doe, DOB 04/12/1985, MRN 88213, was prescribed Metformin 500 mg for diabetes.” The privacy layer (Section IV-C) reduces this to “Patient [PATIENT], DOB [DATE], MRN [MRN], was prescribed Metformin 500 mg for diabetes,” leaving the clinical content untouched. M4 then extracts three structured entities from that sanitized sentence:

```
{ "text": "Metformin", "label": "MEDICATION",
  "start_char": 61, "end_char": 70 }
{ "text": "500 mg", "label": "DOSAGE",
  "start_char": 71, "end_char": 77 }
{ "text": "diabetes", "label": "DISEASE",
  "start_char": 82, "end_char": 90 }
```

Each of these three entities is present in, and only in, the sanitized sentence —no clinical value appears in the extraction output that was not actually in the input text. GLINER and the deterministic rule set jointly cover this example —MEDICATION and DISEASE come from the zero-shot model, while DOSAGE is well suited to the pattern-based rule path —which is representative of how the two mechanisms divide labor across the eight supported categories in the six-document set as a whole (Section VI-E).

G. What This Evaluation Does Not Establish

We list explicitly what this six-document run does not establish, in keeping with the evidence-based approach this paper tries to hold itself to throughout. It does not provide manually annotated ground truth for entity-level precision, recall, or F1 on these documents; the entity and redaction counts reported above are the pipeline’s own operational output, not a measurement against an independent reference. It does not provide an independently validated privacy recall/precision figure; the 484 redactions reported in Section VI-D are recorded detections and anonymization operations, not a confirmed true-positive count. Successful execution on six documents should not be read as evidence of large-scale production scalability or of clinical reliability, and the set itself —two long native-PDF lab reports and four short scanned clinical notes —is small and was not constructed as a statistically representative sample of the document population a deployed system would encounter. A GLINER-based clinical-entity extraction experiment and a related OCR-degradation analysis were carried out earlier in this project as separate internal evaluation artifacts; these experiments were conducted separately from the present system evaluation and are outside the scope of this paper.

VII. DISCUSSION

The results in Section VI demonstrate that heterogeneous healthcare documents —long native-text PDFs and short scanned images alike —can be processed through a single pipeline into structured, de-identified, and traceable clinical output, with the pipeline demonstrating zero recorded failures in processing all six evaluation documents. What makes the pipeline interesting as a system is not any given stage’s peak accuracy, but rather that the stages compose well with one another, and that this composability is reflected in the results: OCR output has retained the layout metadata that downstream traceability requires, privacy processing has performed 484 redactions while retaining 164 clinical entities within the same sanitized text for downstream extraction, and extraction has retained character offsets to enable auditable rather than opaque output. The split between entity categories and document types —lab values and lab tests concentrated within the two long structured PDF reports, and symptoms and anatomical findings concentrated within the four shorter free-text scans —is indicative of the division between deterministic extraction of structured clinical patterns and model-based extraction of free-text clinical concepts intended by the authors, though this finding constitutes a descriptive rather than controlled observation on the behavior of the two mechanisms.

VIII. LIMITATIONS AND FUTURE WORK

The scope of this paper is limited to the M1 through M4 portion of HealthParse —ingestion, OCR and preprocessing, privacy de-identification, and clinical information extraction. The broader project encompasses additional downstream capabilities including document-type classification that are not evaluated, diagrammed, or asserted to be a contribution of this paper; we mention this explicitly so it is not assumed to be in the scope of this paper. Extending from character-offset traceability to full resolution of an entity back to a highlighted region of the original document image is another concrete next step that builds on the OCR layout metadata that has already been preserved in M2. The current M4 configuration does not involve explicit long-document chunking. As a result, documents that exceed the model’s sequence length may be truncated during GLiNER inference while deterministic rules continue to apply to the entire sanitized text.

IX. CONCLUSION

This paper described HealthParse’s ingestion-through-extraction pipeline: document validation and asynchronous processing, dual native-PDF and OCR text extraction with layout fidelity and confidence metadata, Presidio-based privacy de-identification, and hybrid GLiNER-plus-rules clinical entity extraction with character-offset provenance. In addition to the implementation description, we reported an end-to-end evaluation on a six-document mixed-format corpus: 100% observed execution completion on the six-document evaluation set, mean 5.83 seconds of processing time per document, 164 structured clinical entities extracted across eight categories, and 484 recorded privacy redactions across eleven identifier types. The contribution here is the integrated, implemented pipeline and the concrete execution evidence collected against it —not a claim of state-of-the-art accuracy in any single stage, and not a claim that identifying information is guaranteed to be

removed. Future work includes a labeled-ground-truth accuracy evaluation for both de-identification and clinical entity extraction, expansion of the evaluation set beyond the present six documents, extending traceability to full source-region resolution, and, separately from this paper’s scope, the platform’s downstream document-classification capability.

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APPENDIX

The similarity (plagiarism) report and the AI writing report generated for the **HealthParse** paper are attached on the following pages.

- **Similarity report:** overall similarity of 4%, with no integrity flags raised (3 pages).
- **AI writing report:** 0% of the text detected as AI-generated (2 pages).

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Manuscript ID	ICECONF2026-T1-1381
Manuscript type	Artificial Intelligence & Machine Learning
Title	HealthParse: A Modular Pipeline for Privacy-Aware Extraction of Structured Clinical Information from Heterogeneous Healthcare Documents
Authors	DARSHAN S, DHEERAJ NELLORE, AADITYA PARTHA SARATHY, PONMANI S, DEEPAK K
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Current status	Submitted

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Manuscript Details

ICECONF2026-T1-1381

Manuscript Type:

Artificial Intelligence & Machine Learning

Title:

HealthParse: A Modular Pipeline for Privacy-Aware Extraction of Structured Clinical Information from Heterogeneous Healthcare Documents

Abstract:

A hospital intake desk can receive a typeset lab PDF, a faxed discharge summary, and a phone photo of a handwritten prescription in the same hour, and all must become useful clinical data rather than stored files. HealthParse is a document processing pipeline for such mixed document types, with four stages: ingestion and validation, format-aware text extraction (native-text PDFs through direct parsing; scanned or photographed documents through image preprocessing and Tesseract OCR), privacy-aware de-identification of the recovered text, and hybrid clinical information extraction combining a zero-shot named-entity recognition model with deterministic rules for values such as dosages and lab results. Rather than presenting only a conceptual architecture, this paper shows what each stage produces: the data structures, confidence scores, and character-level provenance it retains. An end-to-end evaluation on a six-document corpus of native-text PDFs and scanned images processed all documents successfully, yielding 164 structured clinical entities across eight categories and 484 identifier redactions across eleven identifier types, at a mean of 5.83 seconds per document. These are operational execution results, not an accuracy benchmark against labeled ground truth. The scope is limited to ingestion through extraction, excluding full table-structure recognition, guaranteed removal of all identifying information, and clinical decision support.

Keywords:

Healthcare Document Processing,Optical Character
Recognition,Clinical Named Entity Recognition,ZeroShot
Learning,Medical Information Extraction,Deidentification,Document
Intelligence

Classification:

Artificial intelligence,Machine learning (general),Transformer
models,Image processing,Computer vision,Privacy preservation

Date of Submission:

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